instability, perhaps even oscillation in the population. But once an ESS is achieved it will stay: selection will penalize deviation from it.

To apply this idea to aggression, consider one of Maynard Smith's simplest hypothetical cases. Suppose that there are only two sorts of fighting strategy in a population of a particular species, named *hawk* and *dove*. (The names refer to conventional human usage and have no connection with the habits of the birds from whom the names are derived: doves are in fact rather aggressive birds.) Any individual of our hypothetical population is classifed as a hawk or a dove. Hawks always fight as hard and as unrestrainedly as they can, retreating only when seriously injured. Doves merely threaten in a dignified conventional way, never hurting anybody. If a hawk fights a dove the dove quickly runs away, and so does not get hurt. If a hawk fights a hawk they go on until one of them is seriously injured or dead. If a dove meets a dove nobody gets hurt; they go on posturing at each other for a long time until one of them tires or decides not to bother any more, and therefore backs down. For the time being, we assume that there is no way in which an individual can tell, in advance, whether a particular rival is a hawk or a dove. He only discovers this by fighting him, and he has no memory of past fights with particular individuals to guide him.

Now as a purely arbitrary convention we allot contestants 'points'. Say 50 points for a win, 0 for losing, −100 for being seriously injured, and −10 for wasting time over a long contest. These points can be thought of as being directly convertible into the currency of gene survival. An individual who scores high points, who has a high average 'pay-off', is an individual who leaves many genes behind him in the gene pool. Within broad limits the actual numerical values do not matter for the analysis, but they help us to think about the problem.

The important thing is that we are *not* interested in whether hawks will tend to beat doves when they fight them. We already know the answer to that: hawks will always win. We want to know whether either hawk or dove is an evolutionarily stable strategy. If one of them is an ESS and the other is not, we must expect that the one which is the ESS will evolve. It is theoretically possible for there to be two ESSs. This would be true if, whatever the majority strategy of the population happened to be, whether hawk or dove, the best strategy for any given individual was to follow suit. In this case the population would tend to stick at whichever one of its two stable states it happened to reach first.